

2nd Order ODEs

Solutions to Problem Sheet 1

1. We follow the usual method of solving homogeneous linear ODEs, where we start by proposing a solution of the form $y(x) = Ae^{mx}$ to obtain the auxiliary equation.

- (a) The auxiliary equation is $m^2 - 1 = 0$, which has solutions $m_1 = 1$, $m_2 = -1$. Since m_1 and m_2 are real and distinct, the general solution of the ODE is

$$y(x) = A_1e^x + A_2e^{-x}.$$

- (b) The auxiliary equation is $m^2 + 1 = 0$, which has solutions $m_1 = i$, $m_2 = -i$. Since m_1 and m_2 are complex conjugates, the general solution of the ODE is

$$y(x) = A_1 \cos x + A_2 \sin x.$$

- (c) The auxiliary equation is $m^2 + 8m + 15 = (m + 3)(m + 5) = 0$, which has solutions $m_1 = -3$, $m_2 = -5$. Since m_1 and m_2 are real and distinct, the general solution is

$$y(x) = A_1e^{-3x} + A_2e^{-5x}.$$

- (d) The auxiliary equation is $m^2 + 6m + 13 = (m + 3)^2 + 4 = 0$, which has solutions $m_1 = -3 + 2i$, $m_2 = -3 - 2i$. Since m_1 and m_2 are complex conjugates, the general solution of the ODE is

$$y(x) = e^{-3x}[A_1 \cos(2x) + A_2 \sin(2x)].$$

- (e) The auxiliary equation is $m^2 - 6m + 9 = (m - 3)^2 = 0$, which has solutions $m_1 = m_2 = 3$. Since m_1 and m_2 are real and equal, the general solution of the ODE is

$$y(x) = (A_1 + A_2x)e^{3x}.$$

2. In each case, we solve the homogeneous equation first to find the complementary function, $y_{CF}(x)$. We then propose and solve for a suitable particular integral, $y_{PI}(x)$. The general solution is then $y(x) = y_{CF}(x) + y_{PI}(x)$.

- (a) The auxiliary equation is $m^2 + 4m - 12 = (m + 6)(m - 2) = 0$, which has solutions $m_1 = -6$, $m_2 = 2$. Since m_1 and m_2 are real and distinct, the complementary function is

$$y_{CF}(x) = A_1e^{-6x} + A_2e^{2x}.$$

Since the right-hand side of the ODE is an exponential, we propose $y_{PI}(x) = \alpha e^x$. Substituting into the ODE, we obtain

$$(\alpha + 4\alpha - 12\alpha)e^x = 14e^x \quad \Rightarrow \quad \alpha = -2.$$

Hence, $y_{PI}(x) = -2e^x$. Finally, the general solution is

$$y(x) = A_1e^{-6x} + A_2e^{2x} - 2e^x.$$

- (b) The auxiliary equation is $m^2 + 4m + 3 = (m + 3)(m + 1) = 0$, which has solutions $m_1 = -3$, $m_2 = -1$. Since m_1 and m_2 are real and distinct, the complementary function is

$$y_{CF}(x) = A_1 e^{-3x} + A_2 e^{-x}.$$

Since the right-hand side of the ODE contains trigonometric functions, we propose $y_{PI}(x) = \alpha \cos x + \beta \sin x$. Substituting into the ODE, we obtain

$$\begin{aligned} & -\alpha \cos x - \beta \sin x - 4\alpha \sin x + 4\beta \cos x + 3\alpha \cos x + 3\beta \sin x = 8 \cos x - 6 \sin x, \\ \Rightarrow & (2\alpha + 4\beta) \cos x + (2\beta - 4\alpha) \sin x = 8 \cos x - 6 \sin x. \end{aligned}$$

Hence,

$$\begin{aligned} 2\alpha + 4\beta &= 8, & 2\beta - 4\alpha &= -6. \\ \Rightarrow \alpha &= 2, & \beta &= 1. \\ \Rightarrow y_{PI}(x) &= 2 \cos x + \sin x. \end{aligned}$$

Finally, the general solution is

$$y(x) = A_1 e^{-3x} + A_2 e^{-x} + 2 \cos x + \sin x.$$

- (c) The auxiliary equation is $m^2 - 9 = 0$, which has solutions $m_1 = 3$, $m_2 = -3$. Since m_1 and m_2 are real and distinct, the complementary function is

$$y_{CF}(x) = A_1 e^{3x} + A_2 e^{-3x}.$$

CAREFUL! Since the right-hand side of the ODE appears in the complementary function, we propose $y_{PI}(x) = \alpha x e^{3x}$. This has derivatives

$$y'_{PI}(x) = (1 + 3x)\alpha e^{3x}, \quad y''_{PI}(x) = (6 + 9x)\alpha e^{3x}.$$

Substituting into the ODE,

$$(6 + 9x - 9x)\alpha e^{3x} = 42e^{3x} \Rightarrow 6\alpha = 42 \Rightarrow \alpha = 7.$$

Hence, $y_{PI}(x) = 7xe^{3x}$. Finally, the general solution is

$$y(x) = A_1 e^{3x} + A_2 e^{-3x} + 7xe^{3x}.$$

3. (a) From question 2(a), the general solution to the ODE is

$$y(x) = A_1 e^{-6x} + A_2 e^{2x} - 2e^x.$$

Applying the boundary conditions:

at $x = 0$, $y(0) = A_1 + A_2 - 2 = 1$, i.e. $A_1 + A_2 = 3$.

$y'(x) = -6A_1 e^{-6x} + 2A_2 e^{2x} - 2e^x$, so

$y'(0) = -6A_1 + 2A_2 - 2 = -4$, i.e. $3A_1 - A_2 = 1$.

Solving for A_1 and A_2 , we obtain $A_1 = 1$ and $A_2 = 2$, hence the particular solution is

$$y(x) = e^{-6x} + 2e^{2x} - 2e^x.$$

(b) Since the left-hand side of the ODE is the same as in part (a), the complementary function is

$$y_{\text{CF}}(x) = A_1 e^{-6x} + A_2 e^{2x}.$$

Since the right-hand side of the ODE is a constant, we propose $y_{\text{PI}} = \alpha$. Substituting into the ODE,

$$-12\alpha = 3 \Rightarrow \alpha = -\frac{1}{4}.$$

Hence, the general solution is

$$y(x) = A_1 e^{-6x} + A_2 e^{2x} - \frac{1}{4}.$$

Note, that the boundary condition $y \rightarrow \text{constant}$ as $x \rightarrow \infty$ forces $A_2 = 0$ since $A_2 e^{2x} \rightarrow \infty$ as $x \rightarrow \infty$ for all non-zero A_2 . Using the other condition, we have

$$y(0) = A_1 - \frac{1}{4} = 1 \Rightarrow A_1 = \frac{5}{4}.$$

Finally, the particular solution is

$$y(x) = \frac{1}{4} (5e^{-6x} - 1).$$

4.

$$y' + y - z = e^t, \tag{1}$$

$$-y + z' + z = e^t. \tag{2}$$

We differentiate (1), then use (2) to eliminate z' ,

$$\begin{aligned} y'' + y' - z' &= e^t, \\ \Rightarrow y'' + y' - y + z - e^t &= e^t \Rightarrow y'' + y' - y + z = 2e^t. \end{aligned}$$

Now we use (1) to eliminate z to obtain

$$y'' + y' - y + y' + y - e^t = 2e^t \Rightarrow y'' + 2y' = 3e^t.$$

The auxiliary equation is $m^2 + 2m = 0$, which has solutions $m_1 = 0$, $m_2 = -2$, so the complementary function is

$$y_{\text{CF}}(t) = A_1 + A_2 e^{-2t}.$$

We propose $y_{\text{PI}}(t) = \alpha e^t$, where $\alpha + 2\alpha = 3$, i.e. $\alpha = 1$. Hence, the general solution for y is

$$y(t) = A_1 + A_2 e^{-2t} + e^t.$$

Finally, we use $z = y' + y - e^t$ to obtain

$$\begin{aligned} z(t) &= -2A_2 e^{-2t} + e^t + A_1 + A_2 e^{-2t} + e^t - e^t, \\ &= A_1 - A_2 e^{-2t} + e^t. \end{aligned}$$

5. (a) The auxiliary equation is $m^2 - 2m + 5 = (m - 1)^2 + 4 = 0$, which has solutions $m_1 = 1 + 2i$, $m_2 = 1 - 2i$, so the complementary function is

$$y_{CF}(x) = e^x[A_1 \cos(2x) + A_2 \sin(2x)].$$

For the particular integral, we propose $y_{PI}(x) = \alpha x + \beta$. Substituting into the ODE

$$0 - 2\alpha + 5\alpha x + 5\beta = 5x + 3.$$

Equating coefficients, $5\alpha = 5$, $5\beta - 2\alpha = 3$, i.e. $\alpha = \beta = 1$. Hence, the general solution is

$$y(x) = e^x[A_1 \cos(2x) + A_2 \sin(2x)] + x + 1.$$

- (b) The auxiliary equation is $m^2 + 6m + 9 = (m + 3)^2 = 0$, which has solutions $m_1 = m_2 = -3$, so the complementary function is

$$y_{CF}(t) = (A_1 + A_2 t)e^{-3t}.$$

For the particular integral, we propose $y_{PI}(t) = \alpha e^t + \beta$. Substituting into the ODE,

$$\alpha e^t + 6\alpha e^t + 9\alpha e^t + 9\beta = 32e^t + 9.$$

Equating coefficients, $16\alpha = 32$, $9\beta = 9$, i.e. $\alpha = 2$, $\beta = 1$. Hence, the general solution is

$$y(t) = (A_1 + A_2 t)e^{-3t} + 2e^t + 1.$$

Applying the initial conditions,

$$y(0) = A_1 + 2 + 1 = 0, \quad \text{i.e. } A_1 = -3.$$

$$y'(t) = -3(A_1 + A_2 t)e^{-3t} + A_2 e^{-3t} + 2e^t.$$

so $y'(0) = -3A_1 + A_2 + 2 = 1$, i.e. $A_2 = 3A_1 - 1 = -10$. Hence, the particular solution is

$$y(t) = -(3 + 10t)e^{-3t} + 2e^t + 1.$$

6. The auxiliary equation is $m^2 - 6m + 13 = (m - 3)^2 + 4 = 0$, which has solutions $m_1 = 3 + 2i$, $m_2 = 3 - 2i$, so the complementary function is

$$y_{CF}(x) = e^{3x}[A_1 \cos(2x) + A_2 \sin(2x)].$$

For the particular integral, we propose $y_{PI}(x) = \alpha e^{3x}$. Substituting into the ODE,

$$(9\alpha - 18\alpha + 13\alpha)e^{3x} = 5e^{3x} \Rightarrow 4\alpha = 5 \Rightarrow \alpha = \frac{5}{4}.$$

Hence, the general solution is

$$y(x) = e^{3x}[A_1 \cos(2x) + A_2 \sin(2x)] + \frac{5}{4}e^{3x}.$$

Applying the boundary conditions,

$$y(0) = A_1 + \frac{5}{4} = 1 \Rightarrow A_1 = -\frac{1}{4}.$$

$$y'(x) = e^{3x}[(3A_1 + 2A_2) \cos(2x) + (3A_2 - 2A_1) \sin(2x)] + \frac{15}{4}e^{3x}.$$

Hence, $y'(0) = 3A_1 + 2A_2 + \frac{15}{4} = 7$, i.e. $A_2 = 2$. Finally, the particular solution is

$$y(x) = e^{3x} \left[2 \sin(2x) - \frac{1}{4} \cos(2x) \right] + \frac{5}{4}e^{3x}.$$

7.

$$\frac{dx}{dt} + x + y = 0, \tag{3}$$

$$\frac{dy}{dt} + 3x - y = \sin t. \tag{4}$$

Differentiate (3): $\frac{d^2x}{dt^2} + \frac{dx}{dt} + \frac{dy}{dt} = 0$.

Substitute for $\frac{dy}{dt}$ from (4):

$$\frac{d^2x}{dt^2} + \frac{dx}{dt} + \sin t - 3x + y = 0.$$

Substitute for y from (3) to obtain

$$\frac{d^2x}{dt^2} - 4x = -\sin t.$$

The auxiliary equation is $m^2 - 4 = 0$, which has solutions $m_1 = 2$, $m_2 = -2$, so the complementary function is

$$x_{CF}(t) = A_1e^{2t} + A_2e^{-2t}.$$

For the particular integral, we propose $y_{PI}(t) = \alpha \cos t + \beta \sin t$. Substituting into the ODE,

$$-\alpha \cos t - \beta \sin t - 4\alpha \cos t - 4\beta \sin t = -\sin t.$$

Equating coefficients, $-5\alpha = 0$, $-5\beta = -1$, i.e. $\alpha = 0$, $\beta = \frac{1}{5}$. Hence, the general solution for $x(t)$ is

$$x(t) = A_1e^{2t} + A_2e^{-2t} + \frac{1}{5} \sin t.$$

From (3),

$$\begin{aligned} y(t) &= -x - \frac{dx}{dt}, \\ &= -A_1e^{2t} - A_2e^{-2t} - \frac{1}{5} \sin t - 2A_1e^{2t} + 2A_2e^{-2t} - \frac{1}{5} \cos t, \\ &= -3A_1e^{2t} + A_2e^{-2t} - \frac{1}{5} \sin t - \frac{1}{5} \cos t. \end{aligned}$$

8.

$$\frac{du}{dt} + 9v = 0, \quad (5)$$

$$\frac{dv}{dt} - 4u = 0. \quad (6)$$

Differentiate (5): $\frac{d^2u}{dt^2} + 9\frac{dv}{dt} = 0$.

Substitute for $\frac{dv}{dt}$ using (6) to obtain

$$\frac{d^2u}{dt^2} + 36u = 0.$$

The auxiliary equation is $m^2 + 36 = 0$, which has solutions $m_1 = 6i$, $m_2 = -6i$, so the general solution is

$$u(t) = A_1 \cos(6t) + A_2 \sin(6t).$$

Using (5),

$$\begin{aligned} v(t) &= -\frac{1}{9} \frac{du}{dt} = -\frac{1}{9} [-6A_1 \sin(6t) + 6A_2 \cos(6t)], \\ &= \frac{2}{3} [A_1 \sin(6t) - A_2 \cos(6t)]. \end{aligned}$$

Applying the boundary conditions,

$$u(0) = A_1 = 3, \quad v(0) = -\frac{2}{3}A_2 = -4 \Rightarrow A_2 = 6.$$

Hence, the particular solution to the system of ODEs is

$$u(t) = 3 \cos(6t) + 6 \sin(6t), \quad v(t) = 2 \sin(6t) - 4 \cos(6t).$$

9. Since this is a linear ODE with constant coefficients, we proceed by proposing the solution $y(x) = Ae^{mx}$ as usual. This yields the auxiliary equation

$$m^4 + 4m^3 + 6m^2 + 4m + 1 = (m + 1)^4 = 0,$$

which has solutions $m_1 = m_2 = m_3 = m_4 = -1$. Hence, the general solution is

$$y(t) = (A_1 + A_2t + A_3t^2 + A_4t^3)e^{-t}.$$

10. The auxiliary equation is $m^2 - a^2 = 0$, which has solutions $m_1 = a$, $m_2 = -a$, so the complementary function is

$$y_{CF}(x) = A_1e^{ax} + A_2e^{-ax}.$$

For the particular integral, we propose $y_{PI}(x) = e^{ax}[\alpha \cos(bx) + \beta \sin(bx)]$.

Differentiating,

$$\begin{aligned} y'_{PI}(x) &= e^{ax}[(\alpha a + \beta b) \cos(bx) + (\beta a - \alpha b) \sin(bx)], \\ y''_{PI}(x) &= e^{ax}[(\alpha(a^2 - b^2) + 2\beta ab) \cos(bx) + (\beta(a^2 - b^2) - 2\alpha ab) \sin(bx)] \end{aligned}$$

Substituting into the ODE,

$$e^{ax}[(2\beta ab - \alpha b^2) \cos(bx) - (2\alpha ab + \beta b^2) \sin(bx)] = e^{ax} \cos(bx).$$

Equating coefficients, $2\beta ab - \alpha b^2 = 1$, $-2\alpha ab - \beta b^2 = 0$. Hence,

$$\alpha = -\frac{1}{b^2 + 4a^2}, \quad \beta = \frac{2a}{b(b^2 + 4a^2)}.$$

Finally, the general solution is

$$y(x) = A_1 e^{ax} + A_2 e^{-ax} + e^{ax} \left[\frac{2a}{b(b^2 + 4a^2)} \sin(bx) - \frac{1}{b^2 + 4a^2} \cos(bx) \right].$$